Link2Min: Advancing Digital Communication and Coordination

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***Abstract*—Link2Min, is designed to enhance email management by scheduling meetings with automatically generated Google Meet links, sending out summarized meeting minutes, classifying emails and notifying users to reply to important messages. Its seamless meeting scheduling, efficient email classification using sophisticated algorithms, and timely notifications ensure that users can maintain productivity and organization effortlessly. The platform automatically generates meeting minutes and sends them via email to participants, providing a robust and efficient solution for managing professional communications.**

Keywords—BERT, OAuth, NLP,

I. INTORDUCTION

In today's fast-paced professional environment, effective meeting coordination is crucial for productivity and decision-making. However, the traditional process of scheduling meetings often suffers from inefficiencies due to prolonged back-and-forth email exchanges among participants. These manual interactions not only consume valuable time but also lead to scheduling conflicts and unnecessary delays. The lack of a streamlined scheduling mechanism results in frustration and reduced efficiency, particularly in organizations that rely on frequent collaboration.

Beyond scheduling challenges, meetings frequently conclude without participants having a comprehensive understanding of the topics discussed or the necessary action items. The absence of well-organized summaries or documented minutes contributes to miscommunication, misunderstandings, and reduced accountability. Participants may leave meetings with different interpretations of the key takeaways, leading to inconsistencies in task execution and follow-ups. This issue is particularly prevalent in remote and hybrid work environments, where asynchronous communication further complicates clarity and alignment among team members.

Another pressing challenge in modern communication workflows is the management of emails. With an overwhelming volume of emails received daily, users often struggle to filter through spam, promotional messages, and redundant notifications. This information overload makes it difficult to identify and prioritize critical communications, leading to missed opportunities, delayed responses, and ineffective decision-making. The lack of intelligent automation in email management not only hampers productivity but also increases cognitive load, making it harder for professionals to focus on essential tasks.

To address these challenges, there is a growing need for automation tools that can optimize meeting coordination, enhance communication clarity, and improve email management. Automating the scheduling process can significantly reduce the time and effort required to organize meetings. Additionally, intelligent filtering mechanisms can help prioritize emails, reducing the risk of missing important messages. By integrating these capabilities, organizations can create a more efficient and structured communication framework, leading to increased productivity and better decision-making.

This project provides a robust platform to streamline professional communication and organization. It includes an automated meeting scheduler that allows hosts to effortlessly schedule meetings and automatically send invitations to participants with relevant details. After the meeting, the tool generates and sends meeting minutes, summarizing key points and action items to all participants. Link2Min seamlessly integrates with existing email platforms, offering a cohesive and user-friendly experience that enhances productivity. Additionally, it features an intelligent spam detection system that analyses incoming emails, filters out spam, and prioritizes important communications, ensuring efficient email management.

II. LITERATURE SURVEY

To gain insights into existing research on meeting scheduling, email management and automated documentation, a comprehensive review of papers was conducted. These studies cover various aspect of scheduling automation, email classification, spam detection and meeting minutes generation.

A. *A Survey on Web-based Meeting Scheduling Application*

MengTAST Thalawattha and DU Vidanagama conducted a cross-sectional survey to analyze issues in meeting scheduling. Their study used

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quantitative data collection and descriptive statistical methods to highlight the advantages of automated scheduling systems, such as streamlined coordination by analyzing participants' availability and reducing miscommunication through proper agenda sharing. However, the study also pointed out key challenges, including dependency on technology, the necessity for reliable internet access, and privacy concerns related to sharing availability information.

B. *An Empirical Study on Email Classification Using Supervised Machine Learning in Real Environments*
Wenjuan Li and Weizhi Meng explored email classification using machine learning algorithms such as Decision Trees and Support Vector Machines (SVM). Their empirical study, conducted with over 1,000 users in three different environments, demonstrated high classification accuracy, ranging from 84.2% to 95.7%. The study highlighted the practical applicability of these classifiers in real-world scenarios, leading to improved user acceptance. However, a major drawback is the dynamic nature of spam, requiring continuous retraining of models to maintain accuracy. Additionally, the need for large labeled datasets presents a challenge due to the associated costs and time-consuming data preparation.

C. *Improving Email Response in an Email Management System Using Natural Language Processing-Based Probabilistic Methods*
Abdulkareem Al-Alwani applied Natural Language Processing (NLP) and probabilistic methods to improve email classification and automate email responses. The study demonstrated that NLP-based classification significantly reduces errors and enhances efficiency in sorting and responding to emails. However, implementing sophisticated NLP models introduces additional complexity and requires substantial computational resources, particularly when dealing with large datasets.

D. *A Hybrid Sentiment Analysis Framework for Large Email Data*
Sisi Liu and Ickjai Lee developed a hybrid sentiment analysis framework for email classification using multiple labeling methods and classifiers, including SVM, Naïve Bayes, and K-means clustering. Their study found that SVM and K-means achieved high classification accuracy, making them suitable for large-scale sentiment analysis. Despite these advantages, the study identified challenges such as the complexity of preprocessing unstructured email data and a high neutral classification rate, which may indicate shortcomings in contextual understanding.

E. *Automated Generation of Meeting Minutes Using Deep Learning Techniques*
Megha Manuel, Amritha S. Menon, Anna Kallivayalil, Suzana Isaac, and Dr. Lakshmi K.S. proposed a deep learning-based model for automating the generation of meeting minutes. Their approach incorporated speaker verification, speech-to-text conversion, and text summarization to generate structured meeting summaries efficiently. The model proved to be highly versatile, supporting 119 languages. However, the study reported a decline in accuracy for longer meetings and multilingual conversations. Additionally, speaker verification faced challenges in large groups, affecting its effectiveness.

F. *Spam and Email Detection in Big Data Platforms Using Naïve Bayesian Classifier*
Parth Mahajan, Muskaan Singh, Harpreet Singh, G. Vijaysekaran, and S. Rosi explored the use of a Naïve Bayesian classifier for spam detection on big data platforms. Their study implemented a three-stage approach involving preprocessing, feature extraction, and anomaly detection. The classifier demonstrated computational efficiency and was suitable for real-time applications. However, the study highlighted potential weaknesses, such as susceptibility to spam evasion techniques like obfuscation. Moreover, the model's effectiveness heavily relied on high-quality initial training data, requiring periodic updates to maintain performance.

G. *AutoMin: Leveraging State-of-the-Art Text Summarization Models to Generate Minutes Using Transfer Learning*
Parth Mahajan, Muskaan Singh, and Harpreet Singh introduced **AutoMin**, an AI-driven system that leverages transfer learning for automated meeting minutes generation. Their model incorporated K-Nearest Neighbors (KNN) for classification tasks and achieved high accuracy using state-of-the-art text summarization techniques. However, the study reported limitations such as difficulties in handling long meeting transcripts, high computational resource requirements, and inherent constraints in the KNN model's performance.

H. *Machine Learning Methods for Spam Email Classification*
W.A. Awad and S.M. Elsenofi explored various machine learning approaches for spam email classification. Their study evaluated multiple classifiers, including Naïve Bayes, SVM, k-NN, and Artificial Neural Networks (ANN), with a focus on preprocessing and feature extraction. The results demonstrated that these classifiers effectively handle high-dimensional email data. However, similar to other studies, the authors emphasized the need for frequent retraining due to the evolving nature of spam. Additionally, they pointed out the significant costs associated with

compiling and labeling large datasets required for training these models.

I. *Content-Based Categorization of E-Mail Using Hidden Markov Model Approach*
Shreyansh U. Saraiya and Prof. Nikita Desai applied a Hidden Markov Model (HMM) to categorize emails based on content. The study found that HMM provided robust probabilistic handling of data variability and effectively captured word order relationships, improving email categorization. However, a key limitation was the extensive labeled training data required to optimize performance. Additionally, the model struggled with contextual understanding when similar words appeared in different contexts, affecting classification accuracy.

J. *Meeting Scheduling Across Heterogeneous Calendars and Organizations Utilizing Mobile Devices and Cloud Services*
Varvana Myllériemi, Olli Korjus, Mikko Raatikainen, Terho Norja, and Tomi Männistö developed a framework to facilitate meeting scheduling across different calendars and organizations using mobile and cloud-based services. Their approach ensured high accuracy in schedule coordination, particularly when using classifiers like Naïve Bayes and Rough Sets. The study also explored hybrid system implementations, such as combining Artificial Immune Systems (AIS) with Rough Sets for enhanced accuracy. However, the researchers noted that KNN exhibited poor precision and slower performance. Additionally, spam detection using Rough Sets had lower recall, requiring further improvements.

From the analysis of these papers, Link2Min offers an all-in-one solution for enhancing professional communication and organization. It includes a user-friendly interface that simplifies meeting scheduling, automatically sending detailed invitations to participants. After meetings, it generates concise minutes summarizing key points and action items, and sends them to all attendees. The system integrates seamlessly with existing email platforms, providing a smooth and intuitive user experience to boost productivity. Additionally, it features a smart spam detection system to analyze incoming emails, filter out spam, and highlight important communications, thereby streamlining email management and communication processes.

III. METHODOLOGY

This system is designed to automate email management, meeting scheduling, and meeting minutes generation. It integrates multiple functionalities, including email classification, Google Meet link generation, and automated transcription. Below is a detailed breakdown of the implementation.

A. *Email Inbox Display*

The system retrieves emails from the Gmail API and stores them in an SQLite database. Users can view their emails in a structured table or list format. Each email entry is clickable, allowing users to see detailed information. Additionally, a refresh button is provided to update the email list with the latest inbox contents.

- **Fetches emails** from Gmail API and stores them in an SQLite database.
- **Displays emails** in a structured table or list view.
- **Clickable entries** allow users to access detailed content.
- **Refresh button** updates the inbox with new emails.

B. *Email Classification*

The email classification module processes incoming emails and categorizes them as either spam or ham (non-spam). Spam emails are stored separately, while legitimate emails are forwarded for further processing. Machine learning models like Support Vector Machines (SVM) or Decision Trees are used for classification.

- **Classifies emails** as spam or ham.
- **Spam emails** are moved to a separate database.
- **Ham emails** are further processed for potential meeting requests.
- **Uses machine learning models** to improve accuracy.

The classification results are displayed in the user interface, showing the number of emails identified as spam or ham. Additional features allow users to correct misclassified emails, improving future accuracy. The dashboard provides a visual summary of email filtering.

- **Displays classified email counts** (spam vs. ham).
- **Allows users to review and correct** misclassifications.
- **Provides priority-based categorization** for important emails.
- **Dashboard presents a visual summary** of classification results

C. *Generate Google Meet Link*

Users can schedule meetings by filling out a form with details such as the meeting name, date, time, and participant email addresses. The system uses OAuth authentication to securely access Google APIs and generate a Meet link, which is automatically stored in the database. Meeting invitations are sent to participants via email.

- **Form-based interface** for entering meeting details.

- **Uses OAuth authentication** to securely access Google Meet API.
- **Generates and stores** a Google Meet link.
- **Sends meeting invites** via email to participants.

The system maintains a record of all generated meeting links, allowing users to access them easily. These links are displayed in a table or list format, where each link is clickable, enabling users to join meetings with a single click.

- **Displays previously generated meeting links** for easy access.
- **Clickable links** allow users to join meetings directly.
- **Reduces manual effort** in searching for meeting links.

For emails classified as legitimate meeting requests, the system automatically triggers the Google Meet scheduler. This module generates a meeting link and stores all necessary details. It also integrates with Google Calendar, ensuring participants receive automatic invitations.

- **Identifies meeting-related emails** from classified messages.
- **Automates Google Meet link generation** and storage.
- **Integrates with Google Calendar** for automatic scheduling.
- **Sends invites to participants** via email.

D. *Meeting Minutes Generation*

The system automates the process of generating meeting minutes using speech-to-text conversion and Natural Language Processing (NLP). It captures real-time captions and processes them into a structured format. The generated minutes are displayed to the user for review and can be stored in the database. Participants receive an email containing the final meeting summary.

- **Automated transcription** captures speech in real time.
- **Uses NLP models** (T5 transformer) to structure meeting minutes.
- **Displays generated minutes** for user review and edits.
- **Stores minutes in SQLite database** and **sends via email** to participants.

After the meeting concludes, the system processes the recorded data and generates structured minutes. It organizes the content into sections such as agenda, discussion points, decisions, and action items. The final minutes are stored in a dedicated database and emailed to all participants.

- **Processes meeting data** after scheduling.
- **Uses real-time captions** to extract discussion points.
- **Formats structured meeting minutes** into clear sections.
- **Stores final document** and **sends via email** to attendees.

After a meeting ends, the system processes the recorded captions and summarizes them using NLP techniques. The summary includes sections such as agenda, key discussion points, decisions made, and action items. Users can review and edit the summary before finalizing it. The minutes are then emailed to participants for reference.

- **Uses NLP for summarization** of meeting discussions.
- **Formats the summary into clear sections:**
  - **Agenda**
  - **Key Discussion Points**
  - **Decisions**
  - **Action Items**
- **Allows users to review and edit** before finalizing.
- **Sends final meeting minutes** via email to participants.

IV. FUTURE SCOPE

A. *Advanced Sentiment and Priority Analysis*

Expanding the email classification system beyond spam and ham classification can significantly enhance email management efficiency. By integrating **Natural Language Processing (NLP) models**, the system can analyze **email sentiment** (positive, negative, or neutral) and **urgency levels** to prioritize important messages. This allows users to address critical emails promptly while deprioritizing less significant ones.

- **Sentiment Analysis:**
  - Utilizes NLP models like **BERT or VADER** to detect email tone (e.g., positive, negative, neutral).
  - Helps categorize emails based on emotional intensity, ensuring urgent or concerning messages are addressed quickly.
- **Urgency Detection:**
  - Identifies keywords like “urgent,” “ASAP,” or “critical” to flag high-priority emails.
  - Uses time-sensitive indicators, such as **deadlines or due dates**, to rank emails based on importance.
- **User Interface Improvements:**
  - Emails could be sorted into **priority inboxes**, ensuring time-sensitive matters are immediately visible.

    - Provides a **dashboard view with color-coded urgency labels** (e.g., red for high priority, yellow for medium, and green for low).
- **Custom Rules & Automation:**
    - Users can define **personalized filters** based on keywords, sender priority, or previous interactions.
    - The system could **automatically suggest responses** for high-priority emails, improving efficiency.

B. *Real-Time Meeting Transcription and Translation*

To support diverse and multilingual teams, integrating real-time **speech-to-text transcription and translation** into Google Meet can improve accessibility and collaboration. This functionality enables meeting participants to receive **live transcriptions** and **translated subtitles**, ensuring smooth communication across different languages.

- **Live Transcription:**
    - Uses **speech-to-text models (e.g., Google Speech API, Deepgram, or Whisper)** to convert spoken words into text.
    - Ensures participants can refer back to discussions without relying on memory.
    - **Improves accessibility** for those with hearing impairments.
- **Real-Time Translation:**
    - Integrates **translation APIs (like Google Translate API or OpenAI models)** to provide **instant subtitles** in multiple languages.
    - Helps international teams collaborate seamlessly without language barriers.
- **Meeting Log and Post-Session Availability:**
    - Saves transcriptions for **post-meeting review**, enabling participants to revisit key discussions.
    - **Searchable archives** allow users to find past conversations quickly.
    - Meeting logs can be **linked to the minutes generation module**, ensuring **accurate summaries**.
- **User Interface Enhancements:**
    - Provides an **on-screen subtitle option** during meetings.
    - Enables **language selection per user**, allowing each participant to receive translations in their preferred language.

C. *Personalized Scheduling Recommendations*

An **Machine Learning meeting scheduler** can improve productivity by predicting the **best meeting times** based on user behavior, **past scheduling trends**, and **workload balance**. This system minimizes scheduling conflicts and optimizes team availability.

- **Smart Time Predictions:**
    - Uses **historical meeting data** to learn when users are most available and active.
    - Avoids scheduling meetings during known **low-productivity hours** (e.g., lunchtime or late evening).
    - Recommends **alternative meeting times** if conflicts arise.
- **Adaptive Meeting Format Suggestions:**
    - The system can suggest **alternative formats** if a meeting is unnecessary.
    - For example, if past similar meetings resulted in emails instead of discussions, the system may propose **asynchronous updates** or a **short check-in** instead of a full meeting.
- **Calendar Integration and Auto-Scheduling:**
    - Integrates with Google Calendar to **auto-block available time slots**.
    - Sends **smart reminders** for upcoming meetings based on urgency and participant engagement.
- **User-Customizable Preferences:**
    - Allows users to set **preferred working hours** to prevent meetings outside of comfort zones.
    - Can adjust meeting durations dynamically based on past trends (e.g., if most meetings finish early, future meetings can be scheduled for shorter durations).

D. *Automatic Meeting Agenda and Outcome Summarization*

To enhance meeting productivity, the system can **automatically generate meeting agendas** before the session and provide **post-meeting outcome summaries** based on discussion points.

- **Pre-Meeting Agenda Generation:**
    - Uses **past meeting discussions** and email content to **predict topics** that need discussion.
    - Provides **customizable agenda templates**, allowing users to modify them before the meeting.
    - Highlights key topics based on past **meeting frequency and urgency levels**.
- **Outcome Summarization and Agenda Adherence Tracking:**

- After the meeting, **Natural Language Processing (NLP) models** compare discussions with the planned agenda.
- Identifies **missed agenda points** and ensures action items are recorded.
- Generates a structured **summary**, categorizing content into:
    - **Agenda Adherence:** Tracks which topics were discussed vs. omitted.
    - **Key Discussion Points:** Extracts critical conversations.
    - **Decisions Made:** Highlights final agreements.
    - **Action Items:** Lists responsibilities assigned to participants.

- **Actionable Insights and Follow-Ups:**
    - Identifies pending tasks and **automatically emails** them to responsible participants.
    - Suggests **next meeting topics** based on unresolved issues.
    - Allows **participants to approve/edit summaries** before final distribution.

V. CONCLUSION

The Link2Min project enhances workplace efficiency by integrating intelligent email classification, automated meeting scheduling, and AI-driven meeting documentation. By reducing email clutter and prioritizing critical messages, it ensures users stay focused on important communications without being overwhelmed by spam or low-priority emails.

With automated Google Meet scheduling and smart meeting coordination, Link2Min eliminates the hassle of back-and-forth scheduling, optimizing time management for professionals. Additionally, its real-time transcription and structured meeting minutes generation improve documentation accuracy and streamline follow-ups, ensuring clarity and accountability.

By automating tedious administrative tasks, Link2Min allows professionals to focus on meaningful work, enhance productivity, and improve collaboration. Its seamless integration of AI-driven features makes it a valuable tool for modern professional environments, fostering efficiency and smarter communication.

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